# Lab assignment 2: Social network analysis and modeling

## Challenge-based exercises

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This **Challenge-based exercise** focusses on your extending and assessing your understanding of the content you have covered in Learning-based Exercise. It forms 10% of your course grade, less than the learning-based assignment.
- In Lab Assignment 2, **the Challenge-based exercise are relevant to the Learning-based exercise each week**. In other words, you don't need to wait until the end of Learning-based exercise to start the Challenge-based exercise. Instead, it is recommended that after you finish the Learning-based exercise 1, you can already look into the questions of Challenge-based exercise 1. Same applies to Exercise 2 and 3.
- You will not get input from your teachers, except to clarify the questions. This exercise can be complete during class or outside of class time. Notably, you can get plenty of input from your teachers to help you improve Learning-based Exercise TWO, so prioritise that exercise in class time.
- You should **NOT** work with another pair: each pair should work alone. Very similar answers in different pairs may be seen to reflect plagiarism.
- Each pair should submit a separate answers document specifically for the challenge-based exercise, at the separate submission link on BrigthSpace.

# Exercise 1 (relevant to the exercise 1 in the Learning-based)

### Data preparation

In addition to the Facebook network you have used in the Learning-based exercise, you will use the Highschool network here.

Download the "Highschool_network_att.csv", "Highschool_network_att.csv", and "LargeFacebook_network_att.csv", and "LargeFacebook_network_edge.csv" from BrightSpace. Import the `.csv` data into `R` and build the two networks.

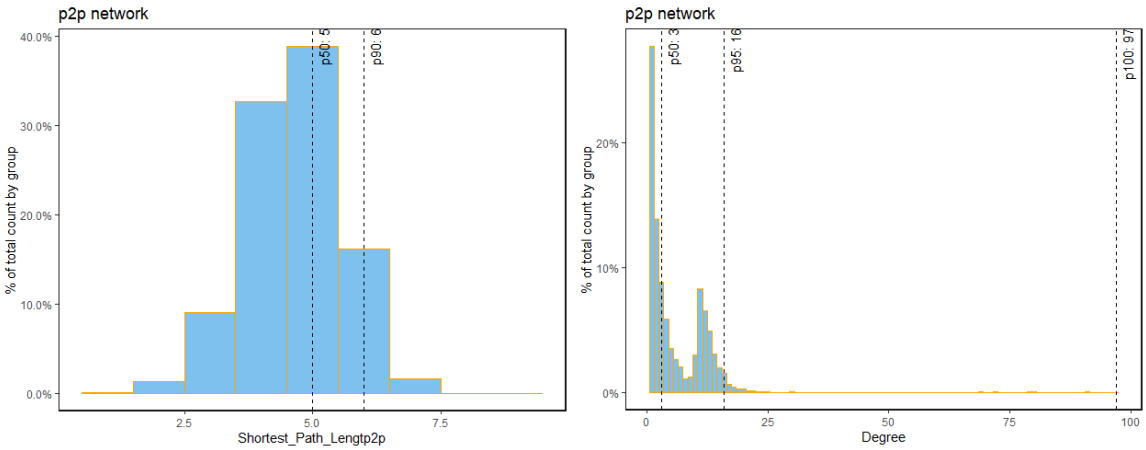
**Question 1 (3 points):**

- For both the Highschool and Facebook networks, calculate the shortest path lengths between every pair of two nodes. How many percentage of nodes can be reached within 6 path lengths? Does "six degree of separation" apply to each network?
- Study the degree distribution of these two networks, are they similar? Then use degree distribution to explain the degree of separation you answered above.

You can calculate the shortest path lengths between every pair of two nodes using *igraph* package:

```
distances(
  Highschool,
  v = V(Highschool),
  to = V(Highschool),
  mode = c("all", "out", "in"),
  weights = NULL) #Shortest path lengths between every pair of two nodes in the network
```

You can consider to use histogram with percentile to study the shortest path lengths and the degree distribution:




(Note: the vertical reference line shows the values of shortest path length or the degree at certain percentiles.)

One important stand of community detection algorithm is based on ***modularity***, which tries to maximize the difference between the actual number of edges in a community and the expected number of edges in the community. However, optimizing modularity in a network is NP-hard, therefore have to use heuristics.

**Question 2 (5 points)** (please submit your code along with your answer)**:**

1) Calculate the modularity of the Highschool network if community is merely identified by a) gender and b) residential hall, respectively.
2) Use the Louvain Community Detection to identify communities in the Highschool network. Compare the modularity value produced by the Louvain algorithm to those in 1),and explain the reasons for the differences.

Codes to customize community and calculate modularity:

```
### customize community by gender ###
genderCommunity<-V(Highschool)$gender
```

```
genderCommunity<-replace(genderCommunity,genderCommunity=="female",1)
genderCommunity<-replace(genderCommunity,genderCommunity=="male",2)
genderCommunity<-replace(genderCommunity,genderCommunity=="unknown",3)
genderCommunity<-as.numeric(genderCommunity)

gender.clustering <- make_clusters(Highschool, membership=genderCommunity)
modularity(gender.clustering)

### Louvain algorithm ###
Louv<-cluster_louvain(graph.object)
modularity(Louv)
```

## Exercise 2 (relevant to the exercise 2 in the Learning-based)

You might realize not every value of $p$ can return you a small-world network that you are looking for. Then a question arises as how can one find the range of $p$. In the Figure 2 of Watts and Strogatz (1998), it explains how can one decide the range of *p* by looking at the dynamics between path length and clustering coefficient.

**Question 3 (4 points):**

1) Start with a regular network of `size=300, nei=6`, first reproduce the Figure 2 of Watts and Strogatz (1998). Then provide the range of p which can turn this regular network (`size=300, nei=6`) into a small-world network.
2) Do you need to rewire significant amount of connections to make the network small-world-like?
3) In the paper of Watts and Strogatz (1998), they pointed out that the value of *p* has two important implications:

*"The idealized construction above reveals the key role of short cuts. It suggests that the small-world phenomenon might be common in sparse networks with many vertices, as even a tiny fraction of short cuts would suffice."*
*"Thus, infectious diseases are predicted to spread much more easily and quickly in a small world; the alarming and less obvious point is how few short cuts are needed to make the world small."*

Use your own words to explain these two implications. For the second implication, connect it with the spread of COVID.